Prerequisites Coding Statement List

- ✅ **Unit 3: Strings**
- ✅ **Unit 4: Sorting**

📘 UNIT 3: STRINGS (20 Problems)

🟢 EASY (10)

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1. Character Recognition UVA 621

🔗 https://onlinejudge.org/external/6/621.pdf
👉 Pattern-based string matching

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2. Decode the Mad Man UVA 10222

🔗 https://onlinejudge.org/external/102/10222.pdf
👉 Character mapping (keyboard shift)

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3. WERTYU UVA 10082

🔗 https://onlinejudge.org/external/100/10082.pdf
👉 Character code manipulation

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4. All in All UVA 10340

🔗 https://onlinejudge.org/external/103/10340.pdf
👉 Subsequence checking

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5. Simply Emirp UVA 10235

🔗 https://onlinejudge.org/external/102/10235.pdf
☞ String reversal + number logic

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6. Mother Bear UVA 10945

🔗 https://onlinejudge.org/external/109/10945.pdf
☞ String cleaning + palindrome

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7. Soundex UVA 10260

🔗 https://onlinejudge.org/external/102/10260.pdf
☞ Encoding characters

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8. Common Permutation UVA 10252

🔗 https://onlinejudge.org/external/102/10252.pdf
☞ Frequency-based string comparison

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9. Reverse and Add UVA 10018

🔗 https://onlinejudge.org/external/100/10018.pdf
☞ String reversal logic

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10. Word Scramble UVA 483

🔗 https://onlinejudge.org/external/4/483.pdf
☞ Word-wise string manipulation

🟡 MEDIUM (5)

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11. Extend to Palindrome UVA 11475

🔗 https://onlinejudge.org/external/114/11475.pdf
☞ Pattern searching (KMP concept)

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12. Power Strings UVA 10298

🔗 https://onlinejudge.org/external/102/10298.pdf
☞ String periodicity

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13. Secret Research UVA 621

🔗 https://onlinejudge.org/external/6/621.pdf
☞ Pattern classification

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14. Period UVA 455

🔗 https://onlinejudge.org/external/4/455.pdf
☞ Pattern repetition

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15. DNA Consensus String UVA 1368

🔗 https://onlinejudge.org/external/13/1368.pdf
☞ Frequency + string construction

● **HARD (5)**

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16. Glass Beads UVA 719

🔗 https://onlinejudge.org/external/7/719.pdf
☞ Lexicographically smallest rotation

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17. String Factoring UVA 11022

🔗 https://onlinejudge.org/external/110/11022.pdf
☞ DP + string compression

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18. Compromise UVA 531

🔗 https://onlinejudge.org/external/5/531.pdf
☞ LCS (string DP)

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19. Prince and Princess UVA 10635

🔗 https://onlinejudge.org/external/106/10635.pdf
☞ LIS + mapping (advanced strings)

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20. Repeated Substitution with Sed UVA 1251

🔗 https://onlinejudge.org/external/12/1251.pdf
☞ String transformation + BFS

📚 UNIT 4: SORTING (20 Problems)

🟢 EASY (10)

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1. Age Sort UVA 11462

🔗 https://onlinejudge.org/external/114/11462.pdf
👉 Counting sort

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2. What is the Median UVA 10107

🔗 https://onlinejudge.org/external/101/10107.pdf
👉 Sorting + median

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3. Sort Sort and Sort UVA 11321

🔗 https://onlinejudge.org/external/113/11321.pdf
👉 Custom comparator

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4. Ecological Bin Packing UVA 102

🔗 https://onlinejudge.org/external/1/102.pdf
👉 Sorting permutations

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5. Relational Operator UVA 11172

🔗 https://onlinejudge.org/external/111/11172.pdf
👉 Basic comparisons

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6. Ordering Tasks UVA 10305

🔗 https://onlinejudge.org/external/103/10305.pdf
☞ Topological sort intro

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7. Little Black Book UVA 450

🔗 https://onlinejudge.org/external/4/450.pdf
☞ Sorting structures

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8. Automate the Grades UVA 11777

🔗 https://onlinejudge.org/external/117/11777.pdf
☞ Sorting scores

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9. Box of Bricks UVA 591

🔗 https://onlinejudge.org/external/5/591.pdf
☞ Sorting + balancing

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10. Stacking Cups UVA 12192

🔗 https://onlinejudge.org/external/121/12192.pdf
☞ Sorting with conditions

🟡 MEDIUM (5)

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11. Flip Sort UVA 10327

🔗 https://onlinejudge.org/external/103/10327.pdf
👉 Inversion count

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12. Ultra-QuickSort UVA 10810

🔗 https://onlinejudge.org/external/108/10810.pdf
👉 Merge sort inversion

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13. DNA Sorting UVA 612

🔗 https://onlinejudge.org/external/6/612.pdf
👉 Sorting with custom metrics

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14. Football UVA 10194

🔗 https://onlinejudge.org/external/101/10194.pdf
👉 Multi-level sorting

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15. Contest Scoreboard UVA 10258

🔗 https://onlinejudge.org/external/102/10258.pdf
👉 Sorting + simulation

● **HARD (5)**

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16. K Smallest Sums UVA 11997

🔗 https://onlinejudge.org/external/119/11997.pdf
☞ Heap + sorting

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17. Frequent Values UVA 11235

🔗 https://onlinejudge.org/external/112/11235.pdf
☞ Sorting + RMQ

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18. GATTACA UVA 11512

🔗 https://onlinejudge.org/external/115/11512.pdf
☞ Suffix array + sorting

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19. The Skyline Problem UVA 105

🔗 https://onlinejudge.org/external/1/105.pdf
☞ Sweep line + sorting

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20. Bridge UVA 10037

🔗 https://onlinejudge.org/external/100/10037.pdf
☞ Greedy + sorting